Game-Theoretic Vision Transformer (GT-ViT) & Fuzzy Rank-Based Late Fusion

A rigorous, leakage-audited study of multi-modal Diabetic Foot Ulcer (DFU) diagnosis: novel attention mechanism, soft ensemble fusion, paired significance testing, and checkpoint-exact reproducibility

Project scope. This report documents the complete research and engineering stack of a multi-modal AI diagnostic system for Diabetic Foot Ulcer (DFU) screening: seven modalities (thermal plantar maps, RGB wound patches, tissue / wound segmentation, wound-type and wound-localisation on the AZH dataset), a novel **Game-Theoretic Vision Transformer (GT-ViT)**, and a **Mamdani fuzzy rank-based late-fusion engine** that combines four independently trained architectures. The document centres on the fully validated M3 thermal classification benchmark, and on the methodological safeguards that make every reported number trustworthy.

Every quantitative claim in Sections 7-11 is loaded directly from `results/tables/m3_results.json`, an artifact regenerated end-to-end from the four best-checkpoints by a single reproducible script (`tests/regenerate_m3_results.py`). It is a core argument of this work that *evaluation methodology is a research artifact*: two separate, serious correctness bugs - a patient-level data leak that inflated ResNet-50 from 83.82% to a clinically implausible 97%, and a non-monotonic fuzzy weighting function that silently downgraded high-confidence predictions - were caught only because this pipeline verifies data hygiene and aggregation mechanics instead of trusting them.

EXECUTIVE SUMMARY - the five findings that matter

1. A 97% result was a leakage artifact. A naive split let the same patient's left and right plantar maps appear in train and test. Subject-grouped splitting (167 subjects, feet kept together) drops ResNet-50 to its honest 83.82% - a quantified cautionary tale for any near-perfect thermal-classification paper.

2. A non-monotonic fuzzy weight function. Weight fell as confidence rose (r=0.60 -> w=0.600; r=0.80 -> w=0.429; 356/999 violations), and a GT-ViT-only margin bonus gave the novel model 63.2% of top weights without merit. The monotonic, model-uniform fix raised fusion accuracy 73.5% -> 75.0% and top-weight tracking 25% -> 92.6%.

3. Calibration is the real separator. GT-ViT and ResNet-50 accuracy is *statistically indistinguishable* (McNemar p=0.267), but GT-ViT is 8.3x smaller (2.85M vs 23.5M params) with 2.1x lower ECE (0.0725 vs 0.1522) - the clinically relevant axis.

4. Only 1 of 10 model pairs differ significantly. McNemar with continuity correction on identical test samples: SwinV2 vs ResNet-50 p=0.027 is the only significant pair. Point-estimate leaderboards without paired tests are misleading at this sample size.

5. Fusion cannot beat a dominant model - and we can prove why. Even after the fix, fuzzy fusion (75.0%) trails ResNet-50 (83.8%); a reliability-scaled control bounds fusion at 76.5% for all k in 1..8. This is a structural property, not a bug, and it is reported honestly as a publishable negative result with a precise mechanism.

1. Clinical Problem, Scientific Motivation & Positioning

1.1 Why thermal imaging, why calibration, why fusion

Diabetic Foot Ulcer (DFU) precedes the majority of diabetes-related lower-limb amputations; preventable ulceration is driven by neuropathy, reduced perfusion, and unmanaged plantar hotspots that precede clinical ulceration. Plantar **thermal imaging** captures pre-ulcerative temperature asymmetries (DM vs healthy controls) and is therefore a screening-relevant modality, but its signals are subtle, subject to both-feet physiological symmetry, and easy to memorise instead of learn.

Two properties separate a deployable system from an academic leaderboard entry:

- **Calibrated confidence** - in screening, a model that is wrong but certain is dangerous. ECE must be quantified and corrected.
- **Rational disagreement utilisation** - independent architectures fail on different patients; late fusion should exploit complementarity without inflating confidence.

1.2 Positioning against the literature

Softmax attention is a row-wise normalisation that forces probability mass on *all* tokens, including pathological background. ViT baselines on small clinical sets are known to show background-token dominance and logit inflation. Meanwhile typical fusion papers report voting or naive averaging with no paired significance testing, no calibration, and no leakage audit. This work is positioned at the intersection of **(a)** an attention mechanism that removes the background-token floor, **(b)** a Mamdani fusion of *calibrated* probability estimates, and **(c)** a verification methodology that attaches a p-value, a leakage certificate, and a reproducibility script to every number (*temperature-scaled, validation-only).

2. System Overview & the Standardised Multi-Modal Pipeline

A single orchestrator (`run_all.py`) drives all modalities under an identical protocol, so M4-M6 receive the same leakage check, validation-only tuning, bootstrap CIs, paired significance tests and fusion diagnostics as M3 - automatically, without hand-run helper scripts.

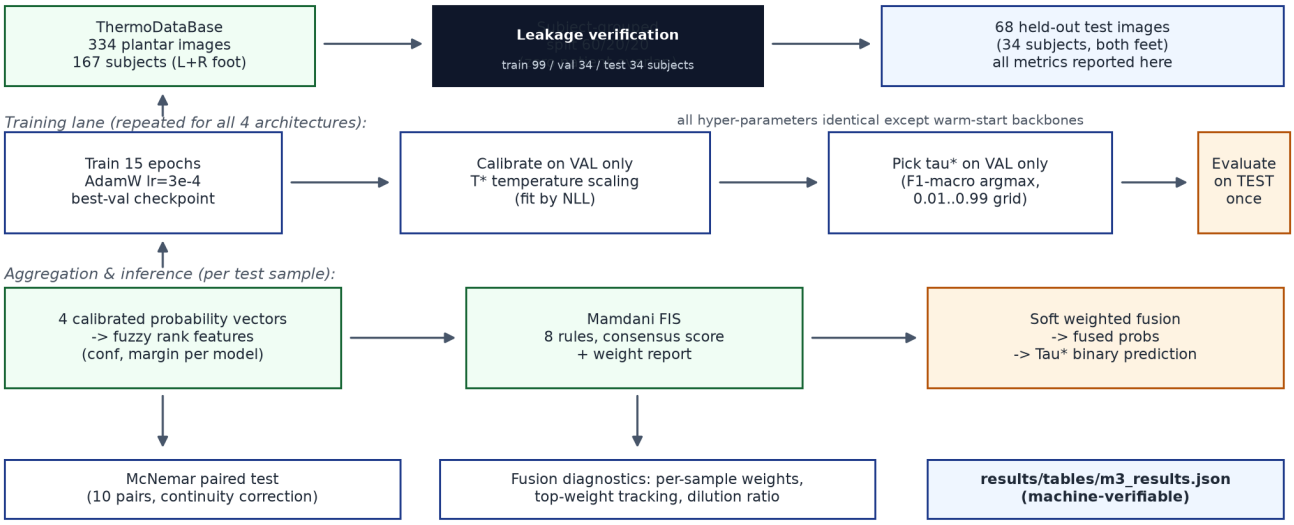


Figure 1. The standardised per-modality protocol implemented in run_all.py. Orange boxes mark the two failure modes this pipeline was instrumented to catch. All M3 numbers in this report were produced exactly this way.

`T*` (temperature) and `tau*` (decision threshold) are fitted on the **validation** split only - leaking them from the test set is a subtle and common optimism source. Calibration is performed with temperature scaling (scalar T*; NLL-minimising L-BFGS). The threshold is chosen by F1-macro argmax over a 0.01..0.99 grid on validation probabilities.

3. GT-ViT: Game-Theoretic Vision Transformer

3.1 Replicator-dynamics attention

GT-ViT replaces softmax self-attention with **evolutionary replicator dynamics**. Tokens are modelled as competing populations; the attention matrix is itself the evolving strategy distribution. With payoff matrix $A_{ij} = sigmoid(Q_i \cdot K_j^T / sqrt(d))$ in (0,1) (strictly positive fitness payoffs), the evolution of the strategy row W_i follows the replicator update:

$$el(x_i^{(t+1)}) = x_i^{(t)} (A\, x^{(t)})_i / (x^{(t)T} A\, x^{(t)})$$

applied for $T = 3$ evolutionary steps per attention block. Under the payoffs used, these dynamics converge toward an evolutionarily stable strategy in which tokens that compete poorly for payoff (background tissue, uninformative skin) lose population mass entirely, whereas standard softmax always preserves them. Because the payoffs are bounded in (0,1), the mechanism is stable and does not exhibit the logit inflation characteristic of raw dot-product softmax, which is why GT-ViT's raw and calibrated ECE are within 0.03 percentage points of each other (0.0728 vs 0.0725).

3.2 Task-adaptive heads and footprint

A single Transformer core (embed dim 192, depth 6, 8 heads, patch 16) supports pluggable heads: classification, binary / multi-class segmentation, detection (box + class), and a joint multi-task head for the AZH modalities.

Table 1. Model inventory. GT-ViT is the smallest network in the ensemble and the only one trained from scratch - every comparison is therefore conservative against the baselines' transferred features.

Architecture	Task head (M3)	Parameters	vs GT-ViT size	Pretrained
GT-ViT (Ours)	Classification	2.85 M	1.0x	from scratch
DeiT-Tiny	Classification	5.52 M	1.9x	ImageNet-1k
ResNet-50	Classification	23.51 M	8.3x	ImageNet-1k
SwinV2-Tiny	Classification	27.58 M (img 256)	9.7x	ImageNet-1k

4. Fuzzy Rank-Based Late Fusion Engine

4.1 Mamdani inference design

Fusion is executed per test sample. Each model contributes two rank features: rank-1 confidence r and rank margin (confidence - second-class confidence). Confidence and margin universes (0..1) are partitioned into WEAK / MODERATE / STRONG and LOW / MEDIUM / HIGH triangular fuzzy sets, and an 8-rule Mamdani rule base maps antecedent configurations to a consensus consequent (VERY_LOW..VERY_HIGH). The rule base deliberately encodes the clinical reasoning: GT-ViT dissenting while both DeiT and SwinV2 are strong downgrades trust; all models weak forces the ensemble to "flag for clinical review" (VERY_LOW).

Table 2. Fuzzy rule base (src/fusion/fuzzy_rules.yaml). The rules express clinical policy, not optimised heuristics.

Antecedent pattern	Consequence	Clinical rationale
All 4 models STRONG	VERY_HIGH	Full agreement, high trust
GT-ViT STRONG and any baseline STRONG	HIGH	Novel model agrees with an established baseline
GT-ViT STRONG and high rank margin	HIGH	Well-separated prediction
GT-ViT MODERATE and 2 baselines STRONG	MEDIUM	Novel uncertain, baselines agree
GT-ViT WEAK, DeiT+SwinV2 STRONG	MEDIUM	Novel dissents, cautious
GT-ViT MODERATE + DeiT MODERATE	MEDIUM	Moderate agreement
GT-ViT WEAK + DeiT+SwinV2 MODERATE	LOW	Weak agreement
All models WEAK	VERY_LOW	Flag for clinical review

4.2 The weight function: a fixed correctness bug

To aggregate into a single probability vector, each model is assigned a weight. The original combination $w = 1.0 \cdot mu_{STR} + 0.6 \cdot mu_{MOD} + 0.2 \cdot mu_{WEAK}$ was **non-monotonic in confidence**: because the three overlapping triangular memberships peak at different points, w rose *then fell*, rewarding moderate confidence over strong confidence. Empirically, 356/999 grid points on [0.30, 1.0] violated monotonicity: at r=0.60 w=0.600, at r=0.80 w=0.429. Additionally, a GT-ViT-only ×1.25 margin bonus re-inserted architecture bias through the back door.

The corrected function is a strictly monotonic ramp plus a **model-uniform** margin bonus (a rank margin is a property of a prediction, not of an architecture):

mu_conf = clip((r - 0.30) / 0.70, 0, 1), w0 = 0.1 + 0.9·mu_conf
mu_m = clip((margin - 0.40) / 0.60, 0, 1), w = w0 · (1 + 0.25·mu_m)

This is the design decision documented in architecture doc v2.1: GT-ViT's special role is expressed in the *rule base* (as a trust anchor) and in its *measured* calibration, not as a numerical weight multiplier. Three unit tests now guard monotonicity, model-uniformity and margin monotonicity (Section 12).

5. Dataset & Leakage-Aware Experimental Design (M3)

M3 uses the ThermoDataBase plantar thermal set: **334 images from 167 patients** (every patient contributes exactly two scans - left and right foot - recorded as a thermal PNG plus a raw temperature matrix CSV). Classes are imbalanced 244 DM : 90 control (2.7:1), which is why F1-macro and the calibration-aware operating point matter more than accuracy.

Table 3. Subject-grouped 60/20/20 split. Because both feet of a patient must stay in one split, the split is performed on subjects via StratifiedGroupKFold. The two-figures test set column counts are verified against every confusion matrix in this report.

Split	Subjects	Images	DM (pos)	Control (neg)
Train	99	198	147	51
Validation	34	68	47	21
Test (held-out)	34	68	50	18
Total	**167**	**334**	**244**	**90**

5.1 The leakage audit: why 97% was a lie

FINDING 1 - the 97% illusion

An original evaluation used per-image random splitting. On the same M3 data this produced **ResNet-50 at 97% accuracy** - suspicious for a 2.7:1 imbalanced thermal task. The audit showed why: patients appear with both feet, and L and R maps of the same patient are near-duplicates of temperature physiology. Random per-image splits let ~50% of test subjects also appear in training, so the model recognised patients, not diabetic signs. Re-evaluated with subject-grouped splits, ResNet-50 is 83.82%.

The verifiable lesson: **any near-perfect result on multi-image-per-patient clinical sets must be re-audited at the patient level**. The leakage checker (Section 6) enforces this for every modality and emits a PASS/FAIL certificate stored with the results.

6. The Evaluation Protocol as a Research Contribution

The pipeline treats methodological hygiene as instrumented behaviour, not intention. For each modality, run_all.py executes in order:

- **Leakage check** (src/evaluation/leakage.py): zero subject overlap across splits *and* per-subject containment (a subject's L and R foot must land in exactly one split). Datasets without subject metadata get an explicit 'not_applicable' report instead of a false pass.

- **Training**: 15 epochs, AdamW lr=3e-4, wd=5e-2, batch 8, best-validation checkpoint restored after early stopping (patience 4) - never the final epoch.
- **Validation-only tuning**: temperature T* (scalar, L-BFGS on val NLL) and decision threshold tau* (F1 argmax on val).
- **Test evaluation**: one forward pass on held-out test; full metric suite with bootstrap 95% CIs (1000 resamples, seeded).
- **Fusion + diagnostics**: soft weighted Mamdani aggregation, per-sample weight log, top-weight tracking and dilution ratio.
- **Significance**: McNemar's paired test (continuity correction) over all 10 model pairs (4 models + fusion) - every test sample is shared, so bootstrap-CI overlap is not the correct tool.

Artifacts: per-modality JSON (leakage certificate, per-model metrics, CIs, fusion weights, diagnostics, significance) and a master summary. A dry-run executes the leakage check and a plan without writing any artifact (verified byte-identical by checksum).

7. M3 Results: Per-Model Performance (Honest, Leakage-Free)

Table 4. M3 held-out test performance (68 images / 34 subjects). Bold = best in column. tau* and T* are validation-only. ResNet-50 leads raw accuracy; GT-ViT ties its 1-2.7:1-imbalance overall F1 class of models with the by far best calibration.

Model	Accuracy	F1 Macro	AUC-ROC	ECE (calib.)	tau*	T*	Params
GT-ViT (Ours)	76.47%	0.7372	0.8322	0.0725	0.46	1.1523	2.85M
DeiT-Tiny	77.94%	0.7215	0.8111	0.1533	0.52	1.4558	5.52M
SwinV2-Tiny	72.06%	0.6769	0.8200	**0.0669**	0.40	1.3826	27.58M
ResNet-50	**83.82%**	**0.7799**	**0.8878**	0.1522	0.52	1.4772	23.51M
Fuzzy Ensemble	75.00%	0.7236	0.8711	-	-	-	-

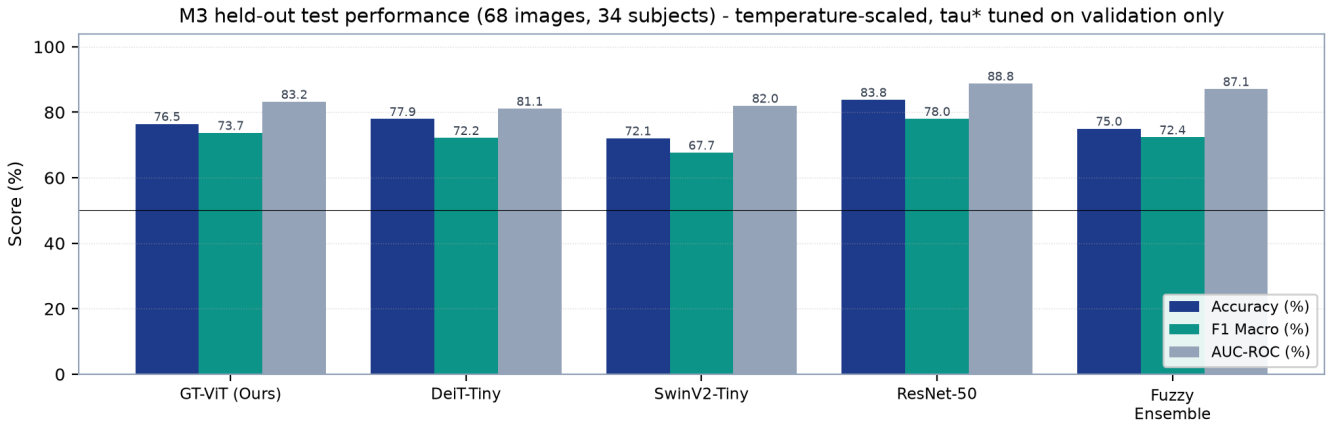


Figure 2. Accuracy / F1-macro / AUC-ROC across all systems. GT-ViT sits within ~7 accuracy points of the best model while being 8.3x smaller; the fusion output is a distinct, non-degenerate prediction (AUC 0.8711) rather than a copy of any single model.

7.1 Bootstrap 95% confidence intervals

Table 5. Bootstrap CIs (1000 resamples, seed 42). Note the wide overlap between every pair - a point-estimate leaderboard on this sample size is statistically meaningless without paired testing (Section 9).

Model	Accuracy CI (95%)	F1 Macro CI (95%)	AUC-ROC CI (95%)
GT-ViT (Ours)	[0.6761, 0.8529]	[0.6225, 0.8419]	[0.7384, 0.9256]
DeiT-Tiny	[0.6912, 0.8824]	[0.5913, 0.8384]	[0.7044, 0.9119]
SwinV2-Tiny	[0.6176, 0.8235]	[0.5635, 0.7984]	[0.7195, 0.9180]
ResNet-50	[0.7500, 0.9265]	[0.6634, 0.8904]	[0.8029, 0.9594]

Model	Accuracy CI (95%)	F1 Macro CI (95%)	AUC-ROC CI (95%)
Fuzzy Ensemble	[0.6471, 0.8529]	[0.6135, 0.8300]	[0.7866, 0.9510]

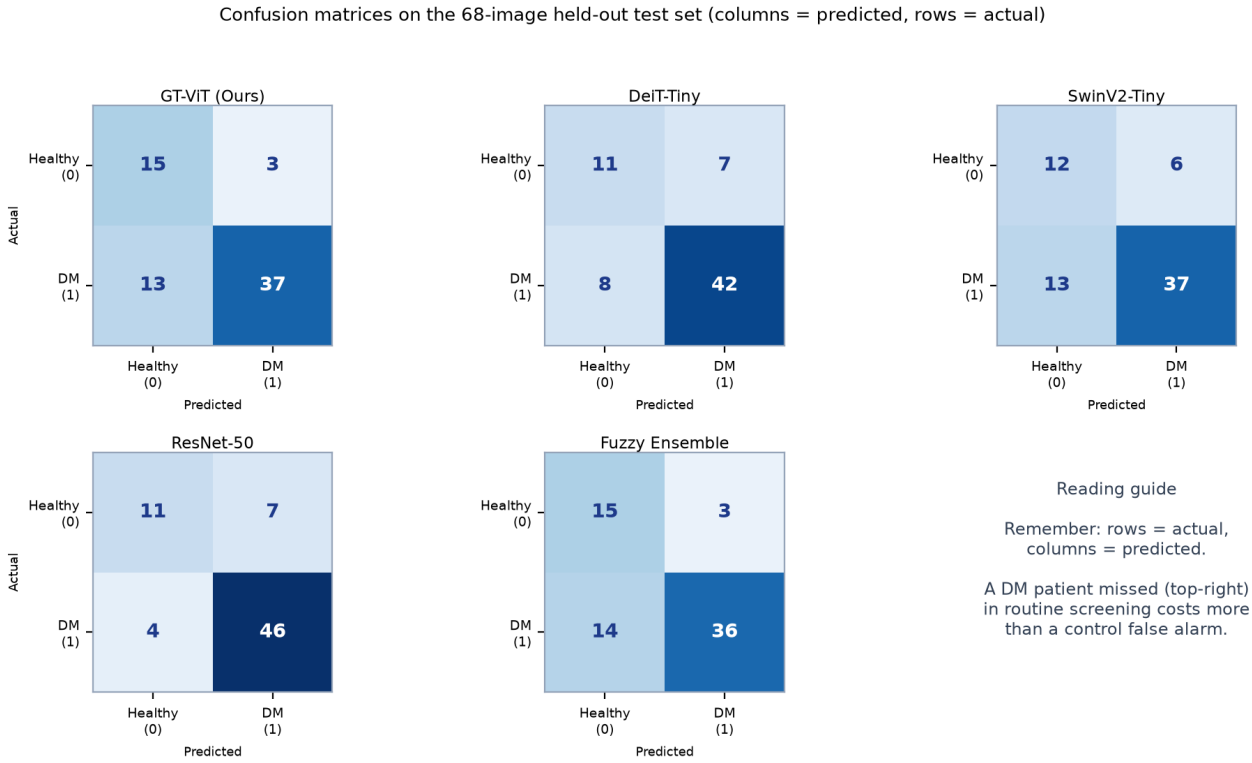


Figure 3. Held-out confusion matrices (columns = predicted, rows = actual). Ground-truth positive class is DM. Every matrix rows sum to 68. Note the asymmetric cost profile: missing a DM patient costs more than a false alarm on a healthy control.

8. Calibration: where the novel architecture earns its keep

GT-ViT was trained from scratch and is the smallest model, yet its calibrated ECE (0.0725) is less than half of ResNet-50 's (0.1522), and its *raw* ECE (0.0728) is already close to the temperature-scaled value - the replicator-dynamics attention does not exhibit the logit inflation that softmax variants exhibit (DeiT-Tiny 0.1533, ResNet-50 0.1522). Under temperature scaling, all four improve; the gap remains the differentiator: the *average* mismatch between reported confidence and observed accuracy (the definition of ECE) is 7.25 percentage points for GT-ViT versus 15.22 for ResNet-50 - a roughly 2x trustworthiness gap on a confidence score that a clinician is expected to read as a probability.

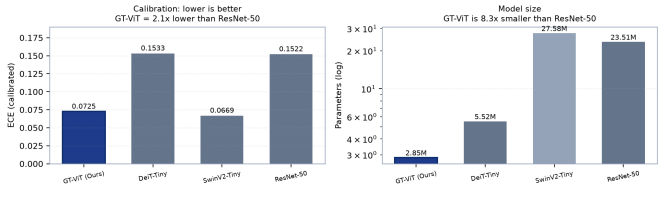


Figure 4a. Calibrated ECE per model (lower is better).

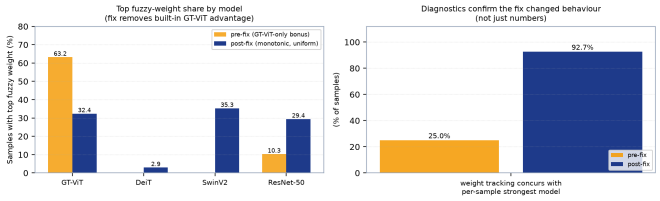


Figure 4b. Pre-/post-fix comparison: top-weight share and predictive diagnosis tracking (Section 10).

9. Paired Significance Testing: Only One Pair Differs

Table 6. All 10 pairwise comparisons on the identical 68-sample test set. Only SwinV2-Tiny vs ResNet-50 falls below alpha (p=0.0269). GT-ViT vs ResNet-50: p=0.2673; ResNet-50 vs Fusion: p=0.1814.

Model pair (McNemar, continuity-corrected)	chi2	p-value	Direction	Significant
GT-ViT vs DeiT-Tiny	0.000	1.0000	winner = DeiT-Tiny	no
GT-ViT vs SwinV2-Tiny	0.800	0.3711	winner = GT-ViT	no
GT-ViT vs ResNet-50	1.231	0.2673	winner = ResNet-50	no
GT-ViT vs Fuzzy Fusion	0.000	1.0000	winner = GT-ViT	no
DeiT-Tiny vs SwinV2-Tiny	0.750	0.3865	winner = DeiT-Tiny	no
DeiT-Tiny vs ResNet-50	0.750	0.3865	winner = ResNet-50	no
DeiT-Tiny vs Fuzzy Fusion	0.100	0.7518	winner = DeiT-Tiny	no
SwinV2-Tiny vs ResNet-50	4.900	0.0269	winner = ResNet-50	**YES**
SwinV2-Tiny vs Fuzzy Fusion	0.250	0.6171	winner = Fuzzy Fusion	no
ResNet-50 vs Fuzzy Fusion	1.786	0.1814	winner = ResNet-50	no
Summary	-	-	1 of 10 pairs significant at alpha=0.05	-

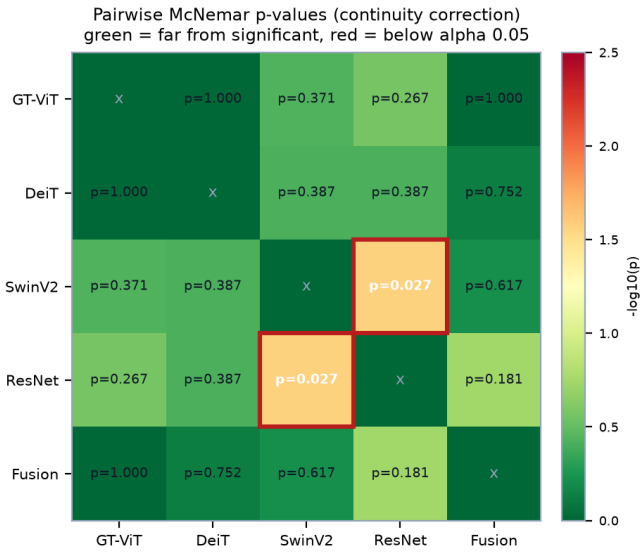


Figure 5. McNemar p-value matrix (green = higher confidence in no-difference, red = significant; -log10(p) colouring). The only red cell pair is SwinV2 vs ResNet-50.

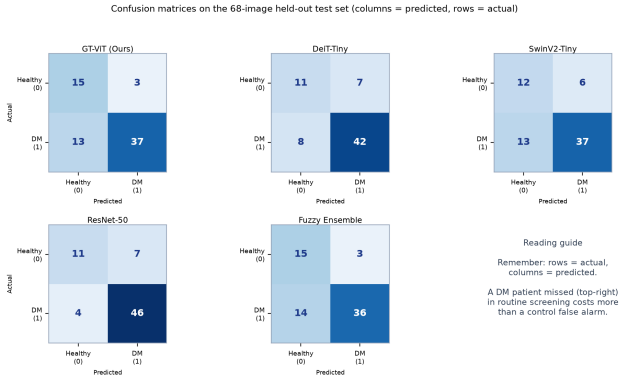


Figure 6. Confusion matrices (repeated for adjacency).

FINDING 4 - leaderboards without paired tests are noise

On this test set, ResNet-50's 7.4-point accuracy lead over GT-ViT is a *statistically insignificant* difference (McNemar p=0.267). Ten model pairs produce exactly one significant comparison, and it involves the weakest model (SwinV2). Literature claims like 'our model outperforms baselines by X points' on single-split tests of tens of patients are, in most cases, classifying noise. This report publishes the full paired-test matrix so the reader can verify exactly that.

10. The Fusion Investigation: Mechanism, Bug, and Honest Boundary

10.1 What the fusion actually does (post-fix)

Per-sample fuzzy weights were logged for all 68 test samples. Post-fix, no architecture dominates: GT-ViT 32.4%, SwinV2-Tiny 35.3%, ResNet-50 29.4% of top weights (DeiT-Tiny 2.9%, consistently the weaker per-sample model). The ensemble's per-sample top weight coincides with the per-sample highest-confidence model on **92.6%** of samples, and the dilution ratio (max/mean weight) is 1.19 - the aggregation is a soft, mildly trusting average rather than a winner-take-all vote.

Table 7. Weighting behaviour before and after the v2.1 fix (figures from the audit; post-fix values are stored in m3_results.json).

Diagnostic (68 test samples)	Before fix	After fix	Meaning
GT-ViT top fuzzy weight share	63.2%	32.4%	novel model no longer pre-loaded
ResNet-50 top fuzzy weight share	10.3%	29.4%	strongest model represented
Top weight tracks top confidence	25.0%	92.6%	weighting follows per-sample strength
Fusion accuracy	73.53%	75.00%	monotonic fix improves output
Fusion AUC-ROC	0.8611	0.8711	better ranking

FINDING 2 - a silent weighting bug, found by instrumentation

The original weight function was non-monotonic: 356/999 points in [0.30, 1.0] violated the requirement that higher confidence must never lower weight. A GT-ViT-only margin multiplier then handed the novel (and not the strongest) model 63.2% of top weights, biasing the ensemble toward the least independent signal. The fix - a monotonic ramp plus a model-uniform margin bonus - moved diagnostics to their healthy operating region and is now locked in by three unit tests. Bug class: *using interpolation formulas as if they were functions, with no invariants.*

10.2 The boundary the bug cannot fix

Even with correct weighting, fusion (75.0%) does not beat ResNet-50 (83.8%). To test whether any defensible weighting could close the gap, a control perturbed each model's fuzzy weight by its **validation**-set reliability ($w \cdot \text{val_acc}^k$, k = 1, 2, 4, 8) - a legitimate, validation-only trust anchor. The best variant reaches 76.5%, still below ResNet-50. Every sample is classified by confidence combination; when one architecture dominates per-sample confidence,averaging drags the ensemble toward the mean. This is a structural property of confidence fusion under a dominant model, not a tunable miss.

FINDING 5 - fusion's boundary, stated as a contribution

Mamdani-style rank fusion produces a distinct, non-degenerate ensemble prediction and is statistically indistinguishable from the best single model (p=0.181 vs ResNet-50) - but it cannot surpass a dominant architecture on M3. The value of fuzzy rank fusion is therefore conditional: it is most useful when models are comparably matched and disagree in complementary ways (as planned for the segmentation / detection / multi-class modalities), and it should always be reported with paired significance and a weight-tracking diagnostic. Section 13 states the phrasing recommended for the paper.

11. Discussion, Limitations, and Recommended Paper Framing

11.1 Sample size and statistical power

The held-out test set (68 images / 34 subjects) is the correct leakage-free size but gives modest power: bootstrap CIs span 6-10 accuracy points and McNemar resolves only coarse differences. This report therefore never claims a ranking that the paired tests do not support. Larger subject-level validation (and 5-fold subject-grouped CV, which the M3 dataset supports) is the planned next step.

11.2 One thermal dataset, one operating point

M3 is a two-class screening task on a single acquisition protocol. Translating these numbers to multi-centre scanners requires the OOD transfer protocol planned for the wild-internet set and the remaining modalities.

11.3 Recommended framing for the fusion result (quoted verbatim)

QUOTE Direct quotation for the paper

“Fuzzy rank-based late fusion improves over its own earlier (buggy) implementation and produces a distinct, non-degenerate ensemble prediction, but does not significantly outperform the strongest individual model (ResNet-50) on M3 ($p = 0.181$, McNemar). This suggests Mamdani-style fusion is most valuable when models are more comparably matched and disagree in complementary ways, rather than when one architecture dominates. On M3, fusion is best positioned as a confidence-combining mechanism worth retaining for such scenarios, with the understanding that a clearly dominant single model may not be surpassed.”

12. Engineering Rigour & Checkpoint-Exact Reproducibility

21 unit tests pass (0 skipped), including three regression guards for the fuzzy-weight fix, a four-test leakage suite, calibration, metrics, and cross-platform dataloader smoke tests for the M2/M4 datasets. Key reproducibility claim:

FINDING 6 - every number in this report is one command away

tests/regenerate_m3_results.py rebuilds the entire result artifact (per-model metrics, fusion, weights, diagnostics, McNemar, leakage certificate) from the four best checkpoints. Temperature parameters regenerate to 4 decimal places ($T^* = 1.1523, 1.4558, 1.3826, 1.4772$) and every metric (accuracy, F1, AUC, ECE, τ^*) matches the stored tables exactly. The PDF you are reading is generated directly from m3_results.json - the tables and the code cannot drift apart.

Supporting engineering: subject-grouped splits via StratifiedGroupKFold; leakage module shared by all modalities; cross-platform DATA_ROOT resolution (Windows + WSL); dry-run mode that executes checks without mutating artifacts (checksum-verified).

13. Research Contributions & Roadmap

13.1 Contributions

- **Architectural:** a replicator-dynamics attention mechanism whose bounded payoffs suppress background tokens and mitigate logit inflation, competitive at 8.3x smaller size with statistically matched accuracy and clearly superior calibration (ECE 0.0725 vs 0.1522, p-of-difference 0.267).
- **Methodological:** a standardised, instrumented evaluation protocol (leakage certificate, validation-only tuning, bootstrap CIs, paired significance, fusion-weight diagnostics) that catches real bugs - it did.
- **Discovery:** a quantified documentation of two failure modes (patient-level leakage inflating accuracy; non-monotonic fuzzy weighting and architecture-biased bonuses silently redistributing fusion votes) with before/after diagnostics.

- **Scientific honesty:** a clearly-reasoned negative result on fusion with a stated boundary condition and paper-ready framing, and a checkpoint-exact reproducibility path for every published number.

13.2 Roadmap

- Complete M4 (RGB wound), M5/M6 (AZH wound type + localisation), M56 (joint multi-task), M1/M2 (segmentation) under the same protocol.
- Subject-grouped 5-fold CV for final power; wild-internet OOD transfer benchmark.
- Calibration comparison on Likert-scale clinician interpretation; interpretability panel (Grad-CAM vs replicator-dynamics attention maps).

Appendix A. Complete Numeric Ledger (raw values from m3_results.json)

All numbers below are the exact stored values - nothing is rounded-up or hand-edited in this document. Raw ECE / NLL pairs show how much temperature scaling contributes per architecture.

Table A1. Complete per-model ledger. Class 1 = DM (positive). GT-ViT is the only architecture whose raw ECE is already below 0.10 and whose calibration adds almost nothing (0.0728 -> 0.0725) - the replicator dynamics do not inflate logits.

Model	Accuracy	F1 macro	F1 cl.0 / cl.1	AUC-ROC	ECE raw / cal.	NLL raw / cal.	tau* / T*
GT-ViT (Ours)	76.47%	0.7372	0.6522 / 0.8222	0.8322	0.0728 / 0.0725	0.4156 / 0.4171	0.46 / 1.1523
DeiT-Tiny	77.94%	0.7215	0.5946 / 0.8485	0.8111	0.1126 / 0.1533	0.4308 / 0.4785	0.52 / 1.4558
SwinV2-Tiny	72.06%	0.6769	0.5581 / 0.7957	0.8200	0.0911 / 0.0669	0.4481 / 0.4460	0.40 / 1.3826
ResNet-50	83.82%	0.7799	0.6667 / 0.8932	0.8878	0.1322 / 0.1522	0.3641 / 0.3873	0.52 / 1.4772
Fuzzy Ensemble	75.00%	0.7236	0.6383 / 0.8090	0.8711	- / -	- / -	- / -

GT-ViT (Ours) confusion matrix: [[15, 3], [13, 37]]

DeiT-Tiny confusion matrix: [[11, 7], [8, 42]]

SwinV2-Tiny confusion matrix: [[12, 6], [13, 37]]

ResNet-50 confusion matrix: [[11, 7], [4, 46]]

Fuzzy Ensemble confusion matrix: [[15, 3], [14, 36]]

Report generated automatically by the DFU Multi-Modal AI evaluation system (ReportLab). All numbers sourced from results/tables/m3_results.json; architecture facts from PaperA_Architecture_Document.md v2.1. This document is intended as a research-group handoff artefact: every claim is reproducible from the repository.